Aidan Vancil

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Work Experience

SpaceX, WA, Redmond, June 2024 - Present.

- **Software Engineer 2**, January 2025 - Present.
  - **Stack/Tools:** Bootchain, Linux, Digital / Analog Communication Protocols, Oscilloscope, State Machines
  - Supported the bring-up of V3 Satellite Propulsion firmware (C++), ground test systems working with PCBAs on unreleased microcontrollers (Python), destructive battery testing at McGregor, and more.
  - Owned software for preliminary ground testing on 7+ major components integrated on the Starlink satellite, contributing to the validation of integrated hardware-software components and reducing defect leakage in downstream testing with no associated satellite contamination incidents in scope.
  - Worked with lead software engineers to root cause a silent, yet crucial clock division issue embedded in abstracted firmware that potentially could have plagued peripheral communications on nearly 65% of all compute nodes on the V3 satellite.
- **Software Engineer 1**, June 2024 - January 2025.
  - **Stack/Tools:** Python, C++, Bash, Docker, Microcontrollers, Low-level Networking, Ansible, Puppet
  - Supported the bring-up of V2 Satellite Propulsion/Lasers firmware, and helped conduct 7+ qualification campaigns to stress-test new PCB designs.
  - Drove performance gains across many test systems, with some PCBA checkout times on boards being reduced up to 70% (i.e., 7 to 2 hours duration) due to expertise with profiling tools and flamegraphs.

Tesla, NV, Reno, January 2024 - May 2024.

- **Software Engineer Intern**, January 2024 - May 2024.
  - **Stack/Tools:** Python, RESTful APIs, ETLs, Kubernetes, Docker, Jenkins, MySQL, Webhooks, Jira, Confluence, Kanban
  - Built 4+ custom developer tools, debugged a multitude of issues with real-time systems over Splunk and Kubernetes, and standardized 5 KPI metrics (FPY, OEE, etc.) across systems factory-wide as tracked through Jira and Github.
  - Piloted the integration/replication of secure S3 buckets (as well as repository templates backed and automated with on-demand pipelines, i.e., Jenkins job .dsl, .yml, Dockerfiles) into the factory's software ecosystem.
  - Spun up an emergency ETL with parallelization within an hour to crunch over 2.5 million records in roughly 4 hours, compared to a prior method's 16-hour runtime.
  - Coordinated a microsoft teams bot that delivered results from a script that processed over 80,000+ records and churned through $1,000,000+ bi-weekly (as measured via Splunk's search engine), with reports getting sent out to over 20+ people of higher-up leadership.

Amazon Web Services (AWS), WA, Seattle, May 2023 - August 2023.

- **Software Engineer Intern**, May 2023 - August 2023.
  - **Stack/Tools:** Python, GraphQL APIs, AWS (S3 + EC2 + RDS + SQS + Lambda), Typescript, Agile, Scrum
  - Implemented 10+ JSON-based IAM policies using access control lists (ACLs) to enforce least-privilege access across AWS services and improve system security.
  - Built and deployed 20+ serverless components using AWS Lambda integrated with SQS, enabling decoupled, event-driven architectures that improved scalability, fault tolerance, and cost efficiency.
  - Collaborated with 8 cross-functional teams to design and deliver system architectures for client-specific use cases, aligning infrastructure decisions with performance, scalability, and reliability requirements.
  - Refactored and owned the rollout of 2+ release branches (Python, Typescript). Used external tools to increase unit test coverage (Istanbul) and deployed new changesets to <10% traffic via canary rollouts, triggering rollbacks using Sentry metrics, if need be.
  - Obtained **AWS Solutions Architect Associate** certificate within a month of tenure.
  - **1st out of 29 participants in Amazon.com internal hackathon**

Education

Bachelor's of Computer Science & Engineering @ The University of Nevada, Reno (UNR): August 2022 - May 2024

~Minor: Mathematics, Dean's List: x4, 4.00 GPA, Magna Cum Laude (ABET Institution)

Associate of Science @ Truckee Meadows Community College, Reno, NV: August 2019 - May 2022

~Summa Cum Laude, Dean's List x4, 3.98 GPA

Software Engineer @ Techwise (Talentsprint) supported by **Google:** March 2022 - August 2023

- Established as 1 of 120 candidates nationwide for a rigorous 18-month software engineering program for high-achieving diverse talent based on leadership potential, academic success, and technical skills. Overseen implementing best coding practices through the mentoring of a Senior Google Software Engineer.

Leadership

Founder/Lead, Google Developer Student Clubs (UNR Chapter), **Google**	August 2023 to August 2024
Vice-President, Nevada Cyber Club	December 2019 to April 2021
Member, UNR, Computer Science Asc. of Machinery	August 2022 to August 2024
Software Engineer Technical Interview Prep (TIP) Fellow, **Edlyft**	February 2023 to August 2023

Technical Skills

- **Languages:** Python, Javascript/Typescript, SQL, C++, Bash, CSS, HTML5, JSON
- **APIs:** RESTful, GraphQL, Insomnia, ReadyAPI, Postman
- **Methodologies:** Agile
- **Security:** Burp Suite, MITRE ATT&CK, Linux, OWASP, OAuth 2.0
- **DevOps:** Docker, Kubernetes (K8s), Ansible, Terraform, Github, Gitlab, Git, Redis, Jenkins, Github Actions
- **Databases:** MongoDB, MySQL, DynamoDB, PostgreSQL
- **Frameworks:** Django, React.js, React Native, Express.js, Next.js, Node.js, Typescript, TailwindCSS, FastAPI, Bootstrap, SCSS
- **Other:** Object-Oriented Programming, Computer Vision (OpenCV), Jira, Atlassian Suite (Confluence/Jira/Stash), UI/UX (Figma), RabbitMQ, Kafka
- **Computing:** Amazon Web Services, Test-Driven Development (TDD), ELK Stack